



Quality and Outcome Measures for Medical Rehabilitation

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SUMMARY PEARLS

- Outcome measures used in the field of rehabilitation are metrics that have been developed and validated to track and report outcomes of clinical care (clinical outcome metrics), clinical care process (process outcome metrics), and functional change (functional outcome metrics).
- Performance measures used in rehabilitation, reported often as a ratio or a percentage, are used to monitor how well the health care system achieves what it purports to achieve, with a focus on quality, cost, efficiency, effectiveness, disparities, safety, and accreditation reporting. Examples include length of stay, readmission rate, mortality rate, and patient satisfaction ratings.
- The American Board of Physical Medicine and Rehabilitation has moved the recertification process from the maintenance of certification to continuing certification, which is now a 5-year cycle.
- In 2013, the Centers for Medicare and Medicaid Services (CMS) developed a national quality strategy (NQS) aligned with three broad aims: better care, affordable care, and healthy people and communities. In 2022, CMS launched the new CMS NQS, a plan aimed at improving the quality and safety of health care for everyone, with a special focus on people from underserved and underresourced communities.
- The evidence-based triad helps clinicians decide on applicability of evidence to individual patients by seeking common ground in the triad of (1) clinical impression based on individual clinical expertise, (2) best available external evidence, and (3) the patient's unique values and expectations.

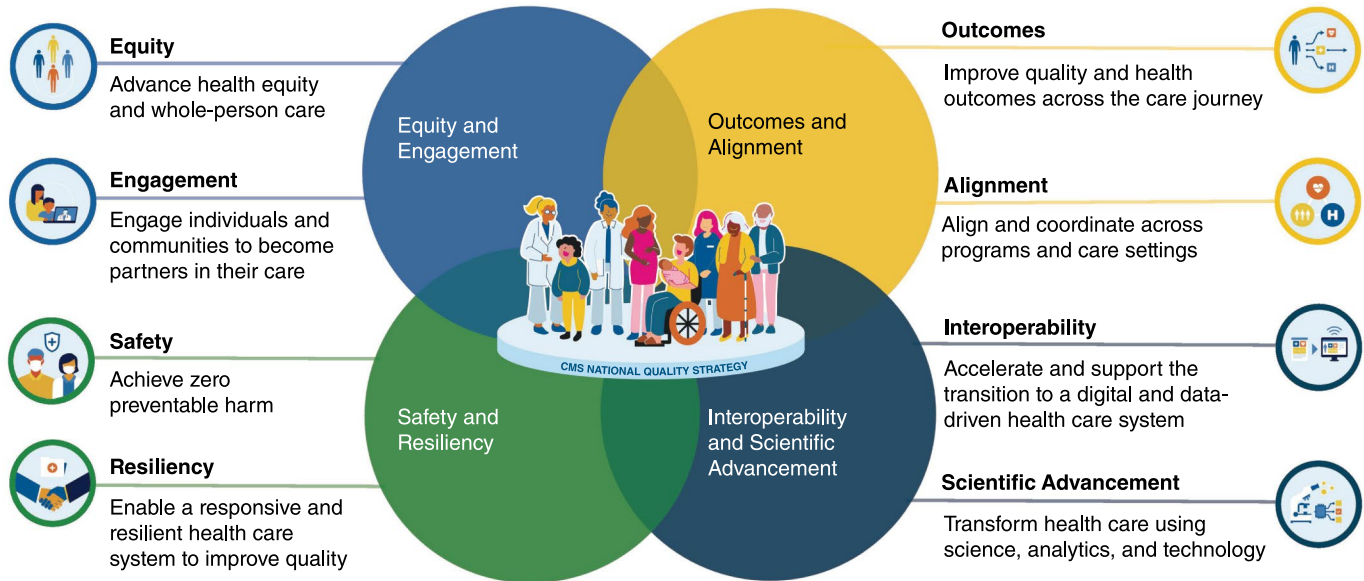
Optimal health care quality can be defined as the provision of safe and effective care that improves outcomes, optimizes health, results in high patient satisfaction, and improves quality of life. As such, the delivery of high-quality care should be a top priority for all rehabilitation professionals. In the United States and worldwide, the evaluation, measurement, and reporting of quality and outcomes have become the business not only of every health care institution and hospital but also of every individual health care provider. This chapter discusses health care quality, quality improvement, evidence-based medicine (EBM), clinical practice guidelines (CPGs), outcome measures, performance measures, practice improvement, patient safety, accreditation, and physician continuing board certification.

Access, affordability, and high-quality health care were the main objectives of the Patient Protection and Affordable Care Act (ACA) of 2010. To accomplish these goals, the Department of Health and Human Services outlined a national quality strategy (NQS). In November 2013, the Centers for Medicare and Medicaid Services (CMS) developed a quality strategy, CMS Quality Strategy 2013–Beyond, which was aligned with the three broad aims of NQS: better care, healthy people/healthy communities, and affordable care. In 2022, CMS launched the new CMS NQS, a plan aimed at improving the quality and safety of health care for everyone, with a special focus on people from underserved and

underresourced communities. CMS's quality strategy goals are depicted in [Fig. 8.1](#). The new CMS NQS has four priority areas, each with two goals. In both versions of CMS NQS, improving safety and quality of care are always the main goals. To achieve these goals, CMS and other health care insurance providers rely on the simple, meaningful, efficient, and accurate measurement of health care quality and safety delivered by providers.

In addition to the CMS, several other governmental and many nongovernmental agencies monitor and oversee health care quality and safety. For example, the National Committee for Quality Assurance (NCQA) is a nonprofit organization focusing on health care quality. Health insurance plans all over the United States seek and obtain the NCQA seal of approval, which reflects their high-quality health care and service. According to the NCQA, its accredited plans cover more than 197 million Americans.⁴² The quality standards and performance measures used by the NCQA and other similar agencies are tools that individual clinicians and organizations use to self-report the quality and safety of health care they have delivered. These measures are then used (1) to compare against local, regional, or national standards; (2) by consumers (patients), media, and other members of the public to compare clinicians, plans, and institutions against each other; and (3) to compare similar scores over time to identify opportunities for improvement, such as designing practice and performance improvement projects.

CMS National Quality Strategy Goals



• **Fig. 8.1** Diagram of the national quality strategy goals of the Centers for Medicare and Medicaid Services (CMS). (From the US Centers for Medicare and Medicaid Services. *Quality in Motion: Acting on the CMS National Quality Strategy*. April 2024. <https://www.cms.gov/medicare/quality/meaningful-measures-initiative/cms-quality-strategy>.)

An important concept in health care quality is that of value. Despite a common objective of improving performance in health care delivery, there are several stakeholders in health care that sometimes have conflicting goals. This can lead to divergent approaches to improving health care quality caused by a lack of clarity and cohesion among stakeholders, resulting in a slow rate of progress. Porter suggests that the achievement of high value should be the overarching goal of health care delivery regardless of one's perspective or role.⁴⁸ Value, in this instance, was defined as health outcomes achieved per dollar spent. Health care outcomes were divided into three tiers: tier 1: health status achieved or retained; tier 2: process of recovery; and tier 3: sustainability of health. Porter recommends that the measurement of “valuable” medical outcomes in any medical condition or disease must include at least one measure in each tier.^{27,48}

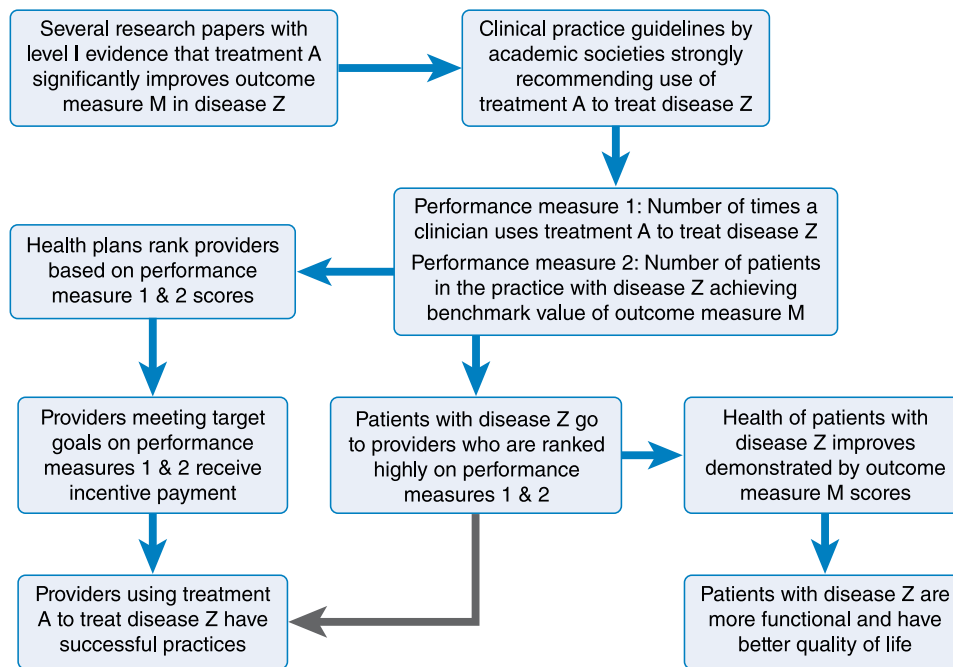
Quality of care is a measure of performance on the six health care aims specified by the Institute of Medicine (IOM): (1) safety, (2) timeliness, (3) effectiveness, (4) equity, (5) efficiency, and (6) patient centeredness. Safety refers to “freedom from accidental injury” and entails reducing errors. An error is the failure of a planned action to be completed as intended or the use of a wrong plan to achieve an aim. Based on performance metrics, providers, individual organizations, and health plans are incentivized to improve quality through accountability, thereby achieving the IOM's six aims.³⁰ Efforts to accurately and efficiently measure quality continue to grow rapidly. These measures are introduced to and adopted by health plans and clinicians based on several factors.

Quality measures need to be meaningful and relevant to patients and society as well as valid and based on sound evidence. The measurement of quality must be transparent, fair, and reliable. Reliability prevents accidents by utilizing design and management principles to reduce conditions that can lead to errors. Transparency supports safety by making the reliability of organizations

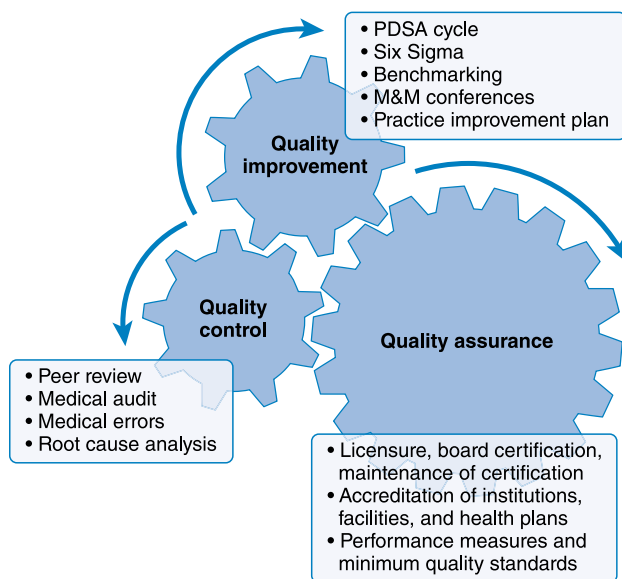
visible by offering comparisons through public reporting of performance measures. Quality measurement must not place too much additional burden on clinicians, their institutions, or health plans, and providing high-quality and safe care must be appropriately rewarded and incentivized. Fig. 8.2 illustrates a model of how evidence-based clinical research translates into quality measures that eventually lead to health improvement. Ideally, all quality measures are based on the highest possible quality of scientific evidence, but some measures are adopted because they are in the best interest of consumers and the society at large even if they are based on best available evidence and not the best possible evidence. The ultimate objective is to deliver high-quality health care and rehabilitation and to improve the health of the population we serve. What one can do to turn the gears of the quality machine depends on the role played in the quality arena. One way is to conceptualize the interplay between how efforts at providing high-quality health care can vary depending on one's perspective and role in the health care system (Fig. 8.3). One example of the effort to improve health care quality is the medical specialty board certification process and continuing certification process.

The Value Equation

Quality measurement dates to Avedis Donabedian's paper²² that categorizes quality into structure, process, and outcomes. Measurement to improve quality began in 1998, when The Joint Commission launched its ORYX initiative, the first national program for the measurement of hospital quality. Subsequently, in 2011, a study was published in the *Journal of Hospital Medicine* based on the study of 4000 hospitals from 2004 to 2008 for performance trends and The Joint Commission association. The study showed that The Joint Commission–accredited hospitals outperformed nonaccredited hospitals in nationally standardized quality measures such as heart failure and pneumonia. But as



• **Fig. 8.2** Schematic illustration of evidence-based quality measures that achieve the desired health outcomes.



• **Fig. 8.3** Health care quality perspectives. *M&M*, Morbidity and mortality; *PDSA*, plan-do-study-act.

these data were plugged into the value equation, it became necessary to ascertain if accreditation signifies higher quality and improvement in quality or if it also helped with performance, thus increasing the value of the care provided.

The value of care is now an important consideration because performance measurement is a complex and expensive process. Therefore the focus of measurement has shifted to accountability measures, which achieve the goal of maximizing health benefits to patients.¹⁸ Accountability measures must meet four criteria:

1. *Research*. There must be randomized controlled trials and more than one study providing scientific evidence to demonstrate

that a given process of care improves health care outcomes (either directly or by reducing the risk of adverse outcomes).

2. *Proximity*. The process being measured must be connected to the outcome it affects.
3. *Accuracy*. The measure should be able to assess whether an evidence-based process is effectively delivered for improvement to occur.
4. *Adverse effects*. The measure should minimize or eliminate unintended adverse effects.

Measures that meet all four criteria are now used for accreditation, public reporting, and pay-for-performance reviews. Performance measurement with these accountability measures has transformed passive payers and consumers into purchasers of efficient high-quality health care, that is value-based performance, where value = quality or outcomes/cost.

Seemingly simple, this equation has a potential to cause confusion if it is not understood that the denominator “cost” refers to the total costs of the full episode of care, not the cost of individual services.^{13,48} Measuring value in the context of improvement in the value of health care by way of performance metrics allows decisions on which services need more spending so that the need for others is reduced. Psychiatrists, regardless of whether practicing inpatient rehabilitation or outpatient musculoskeletal and spine care, need to be aware of the shift in health care’s focus from volume to value-based care.¹³ Today, the higher cost of care is also associated with a poorer quality of care; however, it is imperative to be aware that improvement in value cannot be done by cost containment alone. A cultural change involving innovation and accountability is critical to improving value in health care delivery. Such an improvement is possible with performance measurement because of the transparency with which performance metrics are reported.

Outcome Measures

Broadly speaking, an outcome is the impact of an activity that is driven by input and results in an output (logic model).²⁴ In health

care, an example is a clinical outcome, defined as “a health state of a patient resulting from health care”²¹ by the National Quality Measures Clearinghouse of the Agency of Healthcare Research and Quality (AHRQ). Measuring such outcomes is vital for tracking and refinement purposes and is accomplished by using metrics called outcome measures. For clinical care, clinical outcome measures assess clinical outcomes through qualitative or quantitative means. Outcome measures are selected based on relevance of outcomes to patients, providers, organizations, and health care systems. Examples include positive outcomes such as improved function and negative outcomes such as hospital-acquired skin breakdown. In the health care context, outcome measures can be loosely categorized as clinical outcome measures and process measures.

Clinical Outcome and Process Measures

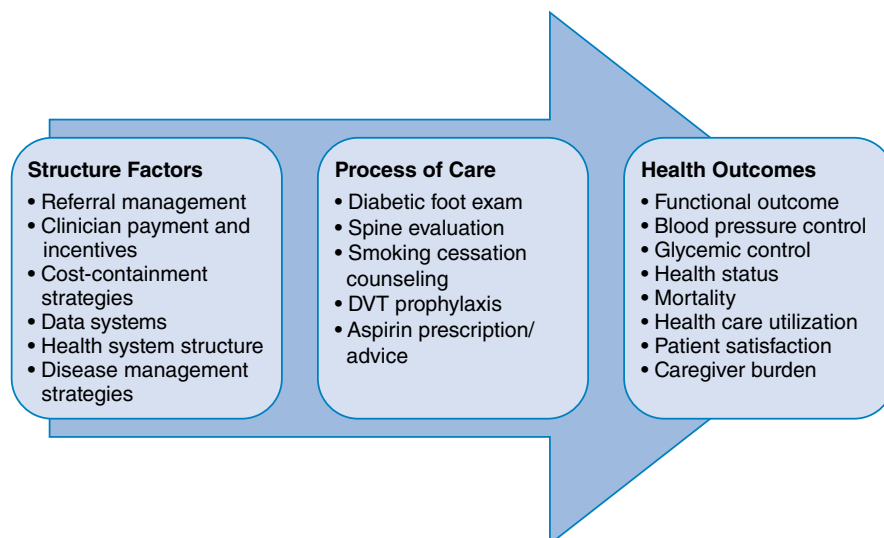
Donabedian conceptualized health care quality evaluation and improvement in 1966 using three dimensions: structure, process, and outcomes.²² Structure is the context in which health care is delivered (e.g., the buildings, equipment, resources, organizational characteristics). The structure of health care is often implicated as an upstream cause of a problem. For example, the previous US health care structure rewarded intervention, and it has been modified to reward value instead to minimize unnecessary interventions. The process of health care is the additive sum of all actions and activities involved in the delivery of health care services. This includes how patients access health care, how patients are assessed, and how diagnoses are made; the way in which treatments are rendered; and the activities involved in counseling, health education, and health promotion. For example, the process of making a diagnosis involves information gathering, conducting an examination, synthesizing data, eliciting additional information in the form of testing or additional consultations, and making an assessment. From Donabedian's perspective, the outcome is the effect of health care on a patient or population. Because it is difficult to ascertain if an outcome is the direct result of health care, it is often challenging to accurately measure if and how much of an outcome is attributable to health care structure or process. Nonetheless, achieving a specific outcome is the reason

specific health care structures are designed and processes take place, and therefore outcomes must be measured. Fig. 8.4 illustrates Donabedian's concept with some rehabilitation-relevant examples included in each dimension.

With regard to the appropriate selection and utilization of an outcome measure, it is important to first determine whether the measure has been evaluated for validity (how well the test measures what it intends to measure) and reliability (the degree to which the measure produces consistent and stable results). Consider the following example: A patient with spinal cord injury is diagnosed with a urinary tract infection (UTI) based on the presence of greater than 100,000 colony-forming units (CFUs) on urine culture. The patient is prescribed a course of antibiotics; on repeat urine culture 2 weeks later, greater than 100,000 CFUs are again found. The goal of the treatment with antibiotics was to eliminate the UTI. Based on the outcome measure of choice in this example—repeat urine culture (a reliable and valid way to measure UTI)—the outcome was not achieved.

The question then becomes why was the outcome not achieved? It could be because of a deficit health care structure, such as poorly functioning equipment that provided incorrect results, or in the process, such as presumptively prescribing an antibiotic and not missing the opportunity to modify it based on culture results. Alternatively, the patient may have taken the prescribed antibiotic for a day and then stopped or may not have been able to afford the copay. Or, as is often the case, a combination of health care factors and patient factors may have affected the outcome. Even in the situation where the patient prematurely stopped taking the prescribed medication, there may have been health care process factors contributing to the poor outcome, such as failure on the part of the provider to explain how to take the medication or failure on the part of the pharmacist to label the medication in the patient's native language. Implementing standard health care processes, such as a telephone or text check with the patient, could have helped to improve patient adherence. As evidenced by this example, even when the management plan is well defined and the evidence behind the treatment is standard of care, achieving a particular health care outcome can be challenging.

Because processes impact outcomes, providers and hospitals measure both processes and outcomes to comprehensively monitor



• Fig. 8.4 Donabedian's concept of health care dimensions and outcomes. DVT, Deep venous thrombosis.

and control both clinical outcomes and fidelity with processes. For example, a rehabilitation unit might be seeking to reduce pulmonary emboli (a negative outcome) and have parameters set for who should be screened using lower extremity Doppler ultrasound. The associated process measure of appropriate screening would be the percentage of those screened among those who met the screening criteria. The higher the percentage, the better. Ideally, all who should have been screened would be, resulting in a 100% score for this measure. Certainly, the rehabilitation unit would also measure the number of patients who had pulmonary embolism (i.e., the clinical outcome). In this case, the goal would be to have zero cases of pulmonary embolism among the at-risk population (the outcome measure). However, the rehabilitation unit leadership might prefer to utilize process measures in addition for several reasons. First, pulmonary embolism is rare, so only a few patients would have pulmonary embolism, and this would not alone reflect the unit's quality of care. Second, the severity and condition type of the patients might be such that the patients on this unit might be at higher risk for pulmonary embolism than those on other units, making a comparison on this outcome measure unfair. Third, the leadership might recognize that the process measure would be much more tangible to the staff and something they could concretely set goals around.

Another measure relevant to the rehabilitation provider is the rate of transfers from an inpatient rehabilitation facility (IRF) unit back to acute care (i.e., the acute care transfer [ACT] rate).⁶⁰ The ACT rate is used as a measure of postacute care quality because these "bounce backs" are common and costly to the health care system,³⁶ especially in the face of vertical integration since the passage of the ACA.⁵³ Just as 30-day readmission rates represent the quality of acute care services, high ACT rates can be representative of inferior postacute care quality because some ACTs could be potentially preventable.⁴¹ A high ACT rate from postacute care or 30-day readmission rate on the acute care side could indicate that the health status of the population served was not stable enough for transferring from acute to postacute care and did not achieve its intended goal of assuring health. However, the reasons why patients are readmitted are many, including some that are well beyond the control of the hospital process during the original admission. In addition, most readmissions may be appropriate. Therefore, when process measure goals are being set for an IRF, it is necessary to consider the underlying reasons for return to the acute hospital from that IRF. Measuring the clinical outcome is hence not enough to make informed changes to the readmissions process. Measures capturing the steps of the care process can ensure a systematic approach to improving outcome and explain why process measures are chosen.

Selecting an appropriate process measure requires strong scientific evidence linking the process to the outcome. As an example, handwashing reduces nosocomial infections; therefore the degree of adherence to hand hygiene policy is an appropriate process measure linked to nosocomial infections. Additionally, there are no obvious contraindications to hand hygiene, so the process measure goal would be 100%. This contrasts with the previous example of ACT rate or 30-day readmission rate, for which a target goal of 0% would be unrealistic and inappropriate. Another important consideration in selecting a process measure is the ease of data collection and analysis. In the era of electronic medical records (EMRs), queries of care processes and steps are much easier to conduct. Examples of EMR process evaluation strategies include keyword searches, categorization by diagnostic codes, order tracking, and provider compliance monitoring. Overall, the

use of process measures in combination with outcome measures helps drive quality improvement in a more systematic fashion than focusing on outcome measures alone. Finally, it is important to adjust for risk factors and/or stratify the population (risk stratification) when using outcome measures by using statistical adjustment of risk techniques. In summary, it is essential to measure both outcomes of health care delivery (outcome measures) and the underlying process (process measures) to ensure that the system is optimized to promote the health of the population served.

Functional Outcome Measures

The International Classification of Functioning, Disability and Health (ICF), released in 2001 as an update by the World Health Organization to their International Classification of Impairments, Disabilities and Handicaps,³³ defines function as an interplay between three levels: at the level of the body part or system (impairment), at the level of the person (activity), and at the level of social role (participation), with influence from an individual's health conditions interacting with the particular environment and personal factors (contextual factors). These environmental and personal factors mediate function at each level in positive and negative ways.³⁵

Functional outcomes⁵⁹ are useful to measure the baseline for a particular level as well as the postintervention change in response to interventions implemented. The goal of an intervention may be to alleviate pain or improve range of motion (body function and structure), to improve walking (an activity), or to improve the patient's ability to work to avoid loss of employment (participation), or a combination of all three. As an example, a knee joint injection may improve the patient's pain, which in turn could improve ambulation and therefore the ability to participate in work. In contrast, using a walker may directly affect ambulation and allow for improved work attendance but does not affect knee function. Recommending worksite adaptations may improve participation but not the patient's body function or ability to ambulate. Because there is not always a clear linear causal relationship between the levels of functioning, there is no guarantee that a prescribed intervention that successfully impacted one level of functioning will positively affect another. Similarly, if contextual factors—namely, environmental and personal factors—are not taken into consideration, functional outcome measures may not accurately reflect the results of health care delivery.

Using the ICF model's functional level as a framework, most medical specialties measure impairments at the body function and structures level, whereas physical medicine and rehabilitation (PM&R) and other rehabilitation disciplines additionally measure activities with assessment instruments, such as the functional independence measure (FIM) and Continuity Assessment Record and Evaluation (CARE) tool/Section GG,⁵⁸ as well as participation with quality of life tools. Functional outcome measures can be further subcategorized in several ways: (1) whether they are reported by the patient, such as the health-related quality of life, or by the provider, such as the FIM; (2) whether they are collected by subjective assessment, such as FIM, or by objective measurement, such as Timed Up and Go test; and (3) whether they can be diagnosis agnostic, such as FIM, or diagnosis specific, such as Amputation Mobility Predictor (AMPPro) for functional Medicare K-level assessment for persons with limb loss.

Functional Independence Measure and CARE Tool/Section GG

In the past, most inpatient rehabilitation programs submitted their patients' FIM scores to the Uniform Data System (UDS). The FIM instrument for adults and the WeeFIM instrument for children measured function in self-care, sphincter control, transfers, locomotion, communication, and social cognition using 18 different items. These items were scored on a 1-to-7 ordinal scale based on help needed from another person to complete the tasks, with a score of 1 indicating total assistance needed and a score of 7 indicating complete independence. UDS provided report cards that detailed FIM change (discharge FIM/admission FIM) and FIM efficiency (FIM change/length of stay) for patients by impairment grouping. UDS also tracked the percentage of patients successfully discharged home. These data were also stratified by impairment type, age, sex, and insurance status. It is important to note that the change in function a population of patients experiences during rehabilitation could be caused by a host of factors outside the control of the rehabilitation physician. Therefore FIM change and FIM efficiency could be considered better outcome measures for the rehabilitation facility than the rehabilitation physician. That being said, there are statistical methods of controlling other factors that would allow for one physician's patients' FIM change and efficiency scores to be compared to other physicians' scores. This would allow for physician-level benchmarking, but it requires a large sample size.

Although much of the rehabilitation literature has utilized the FIM to evaluate functional change until recently, the FIM was replaced by the CARE item/Section GG in October 2019. The CARE item set was developed as a standardized patient assessment tool⁵⁸ for use in postacute settings.⁵⁷ The CARE tool standardizes the medical, functional, cognitive, and social support status assessments for individual patients and includes both core items (assessed on all patients) and supplemental items (based on condition) that assess severity or degree of need. In contrast to the FIM, the CARE tool⁵⁸ measures level of assistance required on a scale of 1 to 6. Similar to the FIM, the CARE tool is used to track individual outcomes, institutional outcomes, and population outcomes with stratification by patient-level factors such as condition type or severity.

Patient-Reported Outcome Metrics

The Patient-Reported Outcomes Measurement Information System (PROMIS) was created by the National Institutes of Health (NIH) to provide several validated, reliable, and precise patient-reported outcome measures.²⁸ These can be used to understand the self-reported health status of patients' physical, mental, and social well-being, especially for chronic disease states and across a wide range of demographics. PROMIS has a core set of questions designed to assess common patient-relevant outcomes across a wide range of conditions and specific item banks that focus on areas such as pain, fatigue, and physical functioning. Depending on the time available and level of detail desired, PROMIS instruments of varying length (e.g., 10, 29, and 57 items) are available.

A particularly attractive feature of PROMIS is the use of item response theory and computerized adaptive testing (CAT). Unlike a traditional questionnaire that is the same for all patients, a CAT instrument uses statistical modeling to determine the most appropriate follow-up question to ask a patient based on response to an initial question, hence tailoring questions specific to each patient.³⁴ If a patient's response to an initial question indicates the ability to complete a task with little help, the CAT selects the next

question with a more challenging task than the first. If that response indicates higher difficulty in completing the more challenging task, the CAT selects the following question with an easier task than in the second question. This process continues until the computer identifies that the preprogrammed confidence interval has been met or the patient has answered the maximal number of items used to estimate the score.³⁴

Pediatric Clinical Outcome, Process, and Functional Outcome Metrics

Similar to adults, pediatric measures can be broadly considered under clinical outcome, process, and functional metrics. The AHRQ has quality indicators² for children receiving care in hospitals, some of which—such as the number of stage III or IV pressure injuries as a secondary diagnosis per 1000 discharges or the number of central venous catheter-related bloodstream infections per 1000—are particularly relevant to rehabilitation care. Similarly, for the rehabilitation population, the outcome of ventilator-associated pneumonia is of particular interest, as hospital complications can lead to worsening debility and a delay in rehabilitative care.

Functional outcomes metrics used for adults may not be a good fit for assessing outcomes for children. The measurement of pediatric functional outcomes should be developmentally appropriate, as self-care, cognitive, and mobility expectations change over the course of childhood. Because children live within the context of their families, it is also important to consider the measurement of family function, health, and well-being. Additionally, patient-reported outcome measures should allow for child reporting as well as parental reporting to assess simultaneous child and parent perspectives.

Choosing Outcome Measures

Choosing an outcome measure is not an exact science because of the plethora of measures available, with varying levels of validation across factors such as diagnosis, age, sex, and care setting. A good starting place for rehabilitation professionals to systematically select outcome measures is the Rehabilitation Measures Database.⁵⁴ This is a publicly available comprehensive collection of evidence-based outcome measures that can be used by clinicians to measure clinical progress for individual patients and by researchers for groups of patients for study design and assessment. Examples of outcome measures in various topic areas within PM&R are shown in Table 8.1.

Major considerations when deciding upon an outcome measure include the nature and utility of the measure for the clinician and the patient, whether the measure has been validated for the specified group of patients, how much of the outcome is directly attributable to the intervention (Box 8.1), and the feasibility of using a metric for the particular application. The latter relates to factors such as ease of learning, ease of administration, and cost. Filters on the website to assist in measure selection include measure type (e.g., observer, patient reported, performance, physiologic, proxy), area of assessment (e.g., mobility, activities of daily living, executive functioning), population (e.g., stroke, spinal injury, osteoarthritis), body part (e.g., chest, lower extremity), and cost.

For example, the use of the 6-minute walk test (6MWT) may be a better indicator of the success of a pulmonary rehabilitation program than the mobility section of the Craig Handicap Assessment

TABLE 8.1 Examples of Outcome Measures in Physical Medicine and Rehabilitation

	Measure and Description	International Classification of Function Domains
Inpatient rehabilitation	FIM: contains 18 items assessing self-care, mobility, bowel and bladder management, cognition, language, and social interaction Section GG / CARE tool: measures level of assistance required on a scale of 1 to 6	Activities/participation
Brain injury	Community Integration Questionnaire: contains 15 items assessing home and social integration and productive activity	Activities/participation
	Ashworth Scale/Modified Ashworth scale: measures spasticity; can be used to assess responsiveness to medication	Body functions/structures
Spinal cord injury	Spinal Cord Independence Measure: assesses self-care, respiratory and sphincter management, and mobility	Body functions/structures Activities/participation
	Spinal Cord Injury Functional Ambulation Inventory: observational gait assessment evaluating parameters of gait, assistive device use, and temporal distance	Activities/participation
Musculoskeletal medicine/pain	Medical Outcomes Study Short Form 36: physical and mental composite domains	Activities/participation
Pediatric rehabilitation	Numeric Pain Rating Scale: 10-point scale; 0 = no pain and 10 = the most intense pain imaginable	Body functions/structures
	WeeFIM: companion measure to the FIM	Activities/participation
	Glasgow Outcome Scale Extended Pediatric Version: used in pediatric traumatic brain injury and evaluates consciousness; independence in and out of the home, school, social, and leisure activities; family and friendships; and return to normal life	Body functions/structures Activities/participation
Stroke	Rivermead Mobility Index: assesses mobility after stroke with 1 direct report and 14 self-observation measures	Activities/participation
	Four Square Step Test: assesses dynamic balance	Activities/participation
Amputation/limb deficiency	10-Meter Walk Test: measures time to walk set distance	Activities/participation
	Medicare K Levels: estimates functional ability and which prosthesis would be most appropriate	Activities/participation
Pulmonary rehabilitation	Saint George Respiratory Questionnaire: addresses pulmonary symptoms, activities, and impact of disease on daily life	Body functions/structures Activities/participation
	2- or 6-Minute Walk Test: measure of endurance	Activities/participation
Spine	Oswestry Disability Index: addresses functional disabilities commonly seen in spine patients	Activities/participation

FIM, Functional independence measure; WeeFIM, FIM instrument for children.

and Reporting Technique (CHART) because there are a multitude of factors that affect CHART, whereas pulmonary rehabilitation is very likely to be a major driver of endurance and speed on the 6MWT. If one of the goals of pulmonary rehabilitation is to improve ambulation speed and endurance, testing clients before and after pulmonary rehabilitation and calculating the average change on the 6MWT of the entire population would be a good measure of the quality and success of the program. The 6MWT is likely feasible in most clinical programs because it is easy to learn and administer, can be assessed by the treating clinician, and can be easily understood by the patient.

Consider another example: Assessment of the outcomes of a pain management program could include a simple calculation of the average change achieved on the 10-point visual pain scale before and after treatment. However, if the goal of the pain program is to improve pain control and affect function such as mobility, use of the 10-point pain scale would be inadequate to determine the program's outcomes. In this case, a functional

outcome measure for mobility could be added. One could also use a functional status metric specific to back pain, such as the back pain functional scale.^{55,56} The ultimate choice of the optimal (combination of) metrics by a given program comes down to clinician, patient, and institutional requirements and preferences. The authors recommend considering a combination of observer- and patient-reported measures to simultaneously assess coherence between clinician and patient perspectives.

Performance Measures and Quality Improvement

Performance measurement is a methodology to monitor how well the health care system achieves what it purports to achieve, with a focus on quality, cost, efficiency, effectiveness, disparities, safety, and accreditation reporting.²⁹ A performance measure is generally a ratio or a percentage that compares providers and institutions

• BOX 8.1 Important Questions to Ask for Determining Which Outcome Measure to Use

- What is my outcome of interest? And to whom is this outcome important?
- What level of function does the intervention intend to address (body functions and structures, activities, and/or participation)?
- For whom will the outcome measure be used (individual patients or groups of patients)?
- Who will be assessed by the outcome measure (a hospital, a practice, a program, a provider)?
- What is the validity of my potential measure of choice?
- What is the reliability of my potential measure of choice?
- Is the outcome measure sufficiently responsive to a change? Is there a ceiling or floor effect that may limit the utility of the measure?
- What is the feasibility of using my potential measure of choice?
- Is the outcome measure patient reported or clinician assessed?
- What are the other factors besides the intervention that could influence the outcome? What number of patients would I need to have to adequately control for those other factors in statistical analyses?

with aggregate benchmark local and national performance. Examples include length of stay, readmission rate, mortality rate, and patient satisfaction ratings.

Performance metrics have helped in transforming health care quality by embedding improvement methodologies within health care and consequently created an accountable data-driven health care organizational culture. Most of these metrics are reflective of collaborative multidisciplinary work involving physicians, nurses, therapists, pharmacists, and other clinicians and nonclinicians. Other stakeholders who drive this culture of quality and patient safety include local and regional purchasers, payers, consumer representatives, and communities.

Performance Measure Development

Development of performance metrics is an exceedingly long and resource-intensive process, requiring financial, human, technologic, educational, and policy resources. Furthermore, adoption of these measures and their impact on care quality improvement are variable. The National Healthcare Quality Report (NHQR) released in May 2014 reported that 70% of recommended quality care was actually received, with wide variations among US states. Although the measures used in the report included a wide range of structure, process, and outcome measures, several important dimensions of quality were not currently measured. Some examples pertaining to rehabilitation medicine include (1) pain reduction and functional improvement in patients undergoing back surgery, total joint replacements, and other orthopedic procedures, and (2) rate of decline in the function of patients with multiple sclerosis. Tracking the rate of change over 4 years of data, the NHQR also reported that whereas the quality of hospital care was improving, ambulatory care was slower to improve. There are few measures to trend limitations in activity, and fewer showed improvement in activity caused by the larger standard errors of estimates of people with activity limitations. Not only was the gap between the best possible care and routinely delivered care substantial, but access to health care also worsened from 2002 to 2010, leveling off in 2011. These statistics from NHQR track health care quality and report everyday experiences of patients and their providers across the nation. The number of people with

disabilities, which is currently 20% of the adult population, is expected to rise with the increase in life expectancy because of the aging of baby boomers. This increase in people born with disabilities and those who acquire disabilities, particularly in the context of multiple chronic conditions and disparities in care, requires the continued development of ways to measure relevant performance. The development of new rehabilitation relevant measures is critical because measures can be retired using criteria such as reaching a 95% threshold and lack of applicability across specialties to remove measures (per CMS).

Considerations for Performance Measure Development

The development of performance measures is difficult for several reasons, which are discussed next.

The Desirable Attributes of a Measure Are Complex

By consensus from the American Medical Association (AMA), The Joint Commission, and NCQA, a performance measure must (1) address a topic area that is of high priority to maximize the health of persons or populations, is financially important, and demonstrates a variation in care and/or potential improvement; (2) have usefulness in improving patient outcomes based on established clinical recommendations, be potentially actionable, and be meaningful to and interpretable by the user; and (3) have a measure design with well-defined specifications, documented reliability, and validity and that allows risk while being feasible, confidential, and publicly available.

Measure Testing Is Elaborate

Measure testing must be conducted in a variety of practice settings. For example, ambulatory measures must be evaluated in solo to large practices. Measure testing requires the involvement of all stakeholders, patients, and providers, including insurers, technology providers, individual or groups of clinicians, and health care organizations.

Measures Need to Meet Multiple Criteria

The AMA's Physician's Consortium for Process Improvement (PCPI)⁴⁷ recommends that performance measures be tested using several criteria, including (1) needs assessment, (2) feasibility and implementation, (3) reliability, (4) validity, (5) unintended consequences, and (6) applications.

Measure Development Life Cycle Is Expensive

The life cycle of a measure, as endorsed by the National Quality Forum,⁴³ an entity that sets priorities for performance measurement nationally, entails (1) studying evidence-based data on disease prevalence, (2) using an assessment tool to study disease severity, (3) using local initiative to measure improvement with process, (4) leading change toward improvement locally, (5) providing public reporting on local performance, (6) endorsing the measure as national consensus standard, (7) retooling the measure for use in electronic health records, and (8) ensuring inclusion in CMS's Meaningful Use program so that the measure adoption is widespread and leads to improvement in patient care.

Risk Adjustment Is Needed to Adapt to Context

Hospitals and provider organizations serve patients with varying risk factors.³⁸ The risk profile can be considered as a combination of patient characteristics, such as age, sex, race, admitting diagnoses,

socioeconomic status, level of education, social support systems, substance use, community or facility from which patients are admitted or sent to the provider, comorbidities, and interactions between multiple diseases and organ systems. Risk adjustment weighs these risk factors to calculate a numeric value that can be compared across patients, provider organizations, and populations, which can be a complicated and long-drawn process.

Performance Reporting Databases Require Both Transparency and Privacy

Public reporting of performance measures involves systems and methods to simultaneously maintain transparency and protect patient privacy.³¹ These measures are reported from electronic health records to public reporting organizations, such as CMS, that maintain large databases of these measures. This process is complex, with multiple stakeholders and cross-cutting measures, making measuring performance a necessary but challenging task.

Strategically Improving Health Care Quality With Performance Measurement and Performance Improvement

Despite challenges, performance measurement is imperative because it helps to provide better quality at a lesser cost. Performance measurement is aligned with the NQS's four priorities with two areas each: <https://www.cms.gov/files/document/quality-motion-cms-national-quality-strategy.pdf> accessed 4/10/2025

1. Promote Aligned and Improved Health Outcomes (Alignment: Align and Coordinate Across Programs and Care Settings; Outcomes: Improve Quality and Health Outcomes Across the Care Journey),
2. Advance Equity and Engagement for All Individuals (Equity: Advance Health Equity and Whole-Person Care; Engagement: Engage Individuals and Communities to Become Partners in Their Care),
3. Ensure Safe and Resilient Health Care Systems (Safety: Achieve Zero Preventable Harm; Resiliency: Enable a Responsive and Resilient Health Care System to Improve Quality) and
4. Accelerate Interoperability and Scientific Innovation (Interoperability: Accelerate and Support the Transition to a Digital and Data-Driven Health Care System; Scientific Advancement: Transform Health Care using Science, Analytics, and Technology).

Over time, there has been a shift from performance improvement being considered valuable for hospitals only, to being considered valuable for providers through relatable evidence-based data for clinicians. This is particularly useful in creating a culture of quality by aligning multidisciplinary teams, comprised of clinicians and nonclinicians, through joint review of evidence-based performance metrics and utilization of that data to work as a team on improving clinical care and processes.

A core guiding strategy underlying performance improvement is reduction in clinical process variation. Clinical variation in patient care processes exists at every level of the US health care system. Electronic health records, computerized physician order entry, and the adverse drug effects alert system are advances achieved through information technology over the last decade to standardize care, prevent errors, and transform the quality of health care services delivered. Documentation of patient care, analysis of care patterns, and institutional- and provider-level feedback of these data have contributed to performance improvement and transparency regarding the quality of care provided, which

consumers now seek, in order to purchase services based on their quality, meaning, and value-based purchasing.²⁷ Within this performance improvement paradigm, performance metrics can be considered as clinical process intelligence to provide decision support, particularly in the management of chronic diseases. Enterprise resource planning is moving to integrate diverse databases such that clinical information will be available across points of care to individuals and communities and to local health care networks, ultimately leading to the evolution of a national network.

Performance Improvement Methodology Overview

Health care performance improvement methodology has been adopted from other industries. Some examples of performance improvement frameworks include the Six Sigma cycle (define, measure, analyze, improve, control,¹⁷ and plan-do-check-act [PDCA]). These performance improvement frameworks place a strong emphasis on identifying and using measures that accurately reflect processes and outcomes of the quality issue.

Performance improvement is a requirement by the Accreditation Council for Graduate Medical Education of all residency programs, including PM&R, under the systems-based practice domain and has been elaborated upon in the latest Clinical Learning Environment Review (v2.0).²⁰ For PM&R residency programs, only a few quality improvement curricula to facilitate such learning are currently discussed in literature²⁶ and show potential for more widespread adoption to improve systems citizenship skill-set facilitation. After residency, the American Board of Physical Medicine and Rehabilitation (ABPMR) requires one performance improvement project to be completed every 5 years for continuing certification. The ABPMR project wizard offers some options for PM&R physicians to complete, or physicians may complete a project of their own design using guideline documents. The performance improvement wizard PDCA methodology sections and set of questions are listed in [Box 8.2](#). There are several ways to categorize quality improvement projects for ease of selection and performance using criteria such as quintuple aims (patient safety/root cause analysis, patient experience, provider well-being, equity), care settings (inpatient, outpatient), and evaluation and management (assessment, interventions, pediatrics, medications). Examples of quality improvement projects under these categories are listed in [Box 8.3](#). Quality improvement projects are described in greater detail in a later section of this chapter.

Policy Implications

Simply using these frameworks to fix problems or create solutions is inadequate.²⁷ A change in thinking and organizational culture to seek and share organized information for patient safety is imperative as we move toward tying payment with quality. As more and more consumers expect evidence for the value of the care they receive, purchasers and payers are using performance measures to ensure quality while simultaneously controlling expenditures on health care with value-based performance. Since 2008, payers such as WellPoint and Aetna have followed in Medicare's footsteps to differential reimbursement based on measurement and comparison. Performance metrics support pay-for-performance initiatives and accreditation in health plans. The American Academy of Physical Medicine and Rehabilitation (AAPM&R) in collaboration with PCPI recently developed measures that are relevant to PM&R. These measures are now available for use in the AAPM&R Registry.^{3,7}

• BOX 8.2 American Board of Physical Medicine and Rehabilitation Quality Improvement Wizard Questions

SECTION 1: General Data

- Background information
- Describe, in detail, your role in the project.
- Dates of your project

SECTION 2: Plan

- What is the problem you are trying to solve?
- What data (objective measurements) do you have that supports this as a problem?
- What is your opportunity statement?
- What is (are) the underlying cause of the performance/quality problem?
- What Institute of Medicine (IOM) Quality Dimensions will be addressed by your project?

SECTION 3: Do

- What change(s) did you implement?

SECTION 4: Study/Check

- Did you achieve your target reported in your opportunity statement?
- Please provide data to support your conclusion.
- If you met your project goal, how will you maintain the success of your project going forward? If you did not quite make your goal, what would you do differently to optimize the success of your project?

Questions quoted verbatim from the American Board of Physical Medicine and Rehabilitation. 8 Steps to Complete a QI Project. <https://www.abpmr.org/CC/QualityImprovement/SelfDirected>. The ABPMR reserves the right to edit, update or alter this information at any time. The ABPMR does not have any agreement with Elsevier, formal or otherwise, to attest to the accuracy of the information provided based on any updates performed following publication of this information.

The Quality Payment Program (QPP)⁶¹ is a legally mandated incentive program for health care quality administered by CMS. Through incentive payments, the QPP rewards attainment of better health care outcomes and higher value by combining two prior quality-reporting programs into one system. There are two paths that providers can choose from to report quality data to CMS: (1) The Merit-Based Incentive Payment System (MIPS)⁴ streamlines four prior independent programs: (a) EHR Incentive Program/Meaningful Use, (b) Physician Quality Reporting System, (c) Value-Based Modifier, and (d) Improvement Activities. Using these components, a composite performance score is determined based on which eligible clinicians may receive a payment bonus, a payment penalty, or no payment adjustment depending on the reporting year. MIPS is intended to ease clinician burden, compared to the prior quality programs, and allows clinicians to choose the activities and measures that are most meaningful to their practice to demonstrate quality. Currently, most eligible clinicians are likely to initially participate in the QPP through MIPS to earn their payment incentive. (2) The second path to report quality data to CMS is the Alternate Payment Model (APM). The APM payment approach rewards high-quality and cost-efficient care. APM can apply to a specific clinical condition, a care episode, or a population. Advanced APMs are a subset of APMs and let practices earn more for taking on some risk related to their patients' outcomes. You may earn additional incentive payment by going further in improving patient care and taking on risk through an Advanced APM. Clinicians who choose Advanced APMs participate in innovative payment models. Currently, only Medicare-focused Advanced APMs are applicable,

• BOX 8.3 Quality Improvement Project Categories With Examples

Categorization Basis: Quintuple Aims

Patient safety/root cause analysis

The "single event" quality improvement project

Patient experience

Improving communication

Provider well-being

Department wellness/burnout initiative quality improvement project/in collaboration with the Association of Academic Physiatrists chair council

Equity

Diversity and inclusion quality improvement project

Categorization Basis: Care Setting

Inpatient

Improving personal protective equipment use

Reducing falls

Reducing pressure injuries

Catheter-associated urinary tract infection management

Reducing inpatient acute care transfers

Inpatient departure checklist

Outpatient

Improving wait times

Telemedicine quality improvement project

Exercise is medicine

Categorization Basis: Evaluation and Management

Assessment

Functional assessment

Interventions

Baclofen pump management

Injection timeouts

Pediatrics

Pediatric fitness surveillance

Cerebral palsy hip surveillance program

Medications

Opioid management: physician drug monitoring program to identify multiple opioid prescriptions and contraindicated drugs

Screening and Opioid Assessment for Patients With Pain-Revised for opioid management

Stroke anticoagulation education

From the American Board of Physical Medicine and Rehabilitation. 8 Steps to Complete a QI Project. <https://www.abpmr.org/CC/QualityImprovement/SelfDirected>. The ABPMR reserves the right to edit, update or alter this information at any time. The ABPMR does not have any agreement with Elsevier, formal or otherwise, to attest to the accuracy of the information provided based on any updates performed following publication of this information.

but in the future, options will also include participation in Medicaid and private payer arrangements.

Pediatric Performance Measures

Pediatric population-specific performance measures are lacking compared with the plethora of measures available for adults. Measures of the Patient-Centered Medical Home (PCMH) now include the word *family* to acknowledge the role of optimal care delivered for children.⁴⁹ The Healthcare Effectiveness Data and Information Set (HEDIS) includes measures on childhood immunization rates, access to primary care providers, height and

weight screening, as well as nutritional counseling and well-child care. Another example of performance measures is the National Expert Panel for the Development of Pediatric Rehabilitation Quality Care Indicators for inpatient rehabilitation management of children with traumatic brain injury, published in 2012.⁵⁰ This measure is classified into the following domains: general management; family-centered care; cognitive-communication, speech, language, and swallowing impairments; gross and fine motor skills impairments; neuropsychological, social, and behavioral impairments; school reentry; and community integration.⁵⁰

Safety and Accreditation

Health plan accreditation standards are used to ensure that they are performing on par with industry standards in quality by measuring their performance. Consumers use these data to compare plans and shop, as do other purchasers, such as employers and state and federal regulators, who utilize these performance metrics so they can pay for performance. The NCQA, Utilization Review Accreditation Commission, and The Joint Commission are such agencies. These agencies rank insurance plans by using tools such as the Medicare Health Outcomes Survey, in which all managed care organizations holding Medicare Advantage contracts are required to enroll in HEDIS. AHRQ's Consumer Assessment of Health Providers and Systems is another such tool.

The Leapfrog Group advocates for quality and safety in hospitals with transparency in pricing and reports on several performance measures. They compare hospitals in terms of preventing medication errors, appropriate intensive care unit staffing, steps to avoid harm, managing serious errors, safety-focused scheduling, and hospital-acquired conditions and provide a Hospital Safety Grade available to consumers on their website.³⁷ Performance metrics data from the CMS are available through its Hospital Compare website.⁴⁰

The accreditation process for inpatient medical rehabilitation facilities through the Commission on the Accreditation of Rehabilitation Facilities (CARF) is internationally recognized as the industry standard for enhancing care in rehabilitation facilities.³⁹ CARF standards are developed through a rigorous process, which includes input from leaders in rehabilitation, stakeholders, and patients served and their families. CARF has defined its mission to "promote quality, value and optimal outcomes of services through their peer review consultative accreditation process and continuous improvement services with the goal of enhancing the lives of persons served." CARF emphasizes the importance of participation of the persons served in their process and utilizes the Universal Stakeholder Participation and Experience Questionnaire to gather input from stakeholders to identify areas that need improvement and drive change in addition to their standards. CARF uses ASPIRE to Excellence as the framework for its standards to measure performance and assess management and quality.¹⁵

Public reporting by CMS on nursing homes and measuring the performance of home- and community-based services with financial incentives linked have been steps toward improving health care quality in the long-term and postacute care arenas. Physician performance metrics are tied into financial incentives. The public reporting of performance metrics is a driver of health care quality because it encourages transparency and accountability.

Evidence-Based Medicine

EBM practice is considered an integral part of providing quality and cost-effective care.²¹ EBM has been defined as "the integration

of best research evidence with clinical expertise and patient values."⁵¹ Utilizing EBM, physicians can address clinical questions by assessing the quality of evidence in medical information available in both basic science and clinical research.⁶³ Sackett et al.⁵² also explained EBM as the integration of individuals' clinical expertise with the best available external evidence from systematic research and recommended using EBM as "the conscientious, explicit, and judicious use of current best evidence in making decisions about the care of individual patients."

It is important to acknowledge the limitations of making clinical decisions solely on one's knowledge of pathophysiology of disease processes, clinical intuition, and individual clinical experience. With the EBM approach, practitioners can use individual clinical knowledge and acumen, careful history taking, and physical examination skills before evaluating existing evidence and combining it with the clinical information obtained in a systematic and reproducible manner.²³ Clinical decisions made using the EBM approach are more reproducible and freer of bias than the traditional approach. Another advantage to this approach is that it allows a clinician to adapt to and accept new scientific evidence more easily. The clinician is enabled to apply the new evidence at the appropriate point in the medical decision-making algorithm to make corresponding changes to one's clinical practice. Equipped with the best new evidence, the clinician can tailor a treatment plan that aligns also with the patient's individual goals and values using a shared decision-making process with the patient.

Assessing, Evaluating, and Applying Evidence

Sackett et al.⁵¹ recommend application of EBM using a simple and straightforward approach to answer any clinical question by (1) defining a clinical question, (2) finding the evidence that will help to answer the question, (3) assessing whether this evidence is valid and important, and (4) applying the evidence to the patient.

For PM&R clinicians trying to apply available scientific evidence to their practice, this means that they would first have to be aware of the existing evidence base in the area of interest. In topic areas where there are few published research articles, this task can be accomplished by performing a basic literature search and assessing the evidence base of the intervention in the articles reviewed. However, as is often the case, several published articles may exist in the topic area of interest, frequently with conflicting results. In such an instance, a review article published in this area often reveals a summary of evidence collated from published research in the area of interest.

In general, the level of evidence provided by a published article can be determined from the level-of-evidence hierarchy (Fig. 8.5). However, one can also appraise the level of evidence of a research article using evidence tables. The Oxford Center for Evidence-Based Medicine (OCEBM) has an easy-to-use table that can help to determine the level of evidence of a published article.⁴⁵ The OCEBM is widely accepted and well regarded in the scientific community. The table adopted by the AAPM&R has five levels for each of four types of studies: therapeutic, prognostic, diagnostic, and economic/decision modeling.⁶⁴ Using either table, reviewers can evaluate the level of evidence of the articles they review to research a clinical question. Once aware of the evidence, the clinician would then have to "conscientiously, explicitly, and judiciously" apply such evidence in making clinical decisions about the care of individual patients.

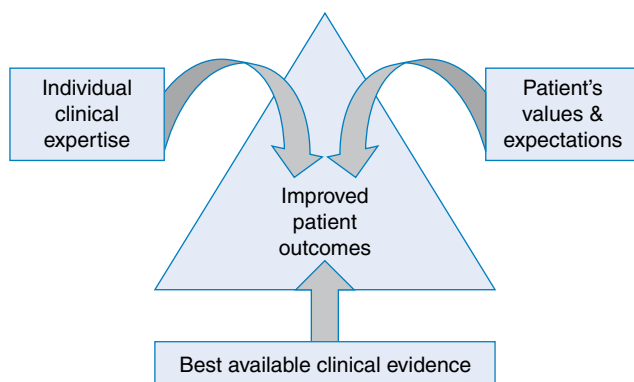


• Fig. 8.5 Evidence-based research pyramid.

For a PM&R clinician reading an original research article or a review article and wondering how it might alter the management of a specific patient, another potentially useful EBM concept is the evidence-based triad.²⁵ This concept provides a guide for deciding on the applicability of evidence to individual patients by seeking common ground in the triad of (1) clinical impression based on individual clinical expertise, (2) best available external evidence, and (3) the patient's unique values and expectations (Fig. 8.6).

When it comes to using EBM to answer a clinical diagnostic decision-making question, the following stepwise approach has been recommended⁶⁵: (1) Define a clinical question and its four components: patient, intervention, comparison, and outcome. (2) Find the available evidence that will help answer the question. (3) Assess whether this evidence is valid and important. (4) Apply the evidence to the patient. This last step includes assessing whether the test can be used, determining if it will help the patient, finding whether the study patients are similar to the patient in question, determining a pretest probability, and deciding if the test will change the management of the patient (Fig. 8.7).

It is recognized that the key steps of EBM practice require some training. EBM has been adopted as part of PM&R residency training over the past 20 years. Annaswamy et al.¹⁴ reported the results of two surveys of PM&R residency programs by the



• Fig. 8.6 Evidence-based triad.

Association of Academic Physiatrists (AAP), which revealed that most programs included training in EBM mostly in the form of pedagogic teaching and journal clubs. Though most of the programs included some EBM training, many do not provide a structured curriculum. The AAP and Cochrane Rehabilitation are collaborating on a road map and curriculum recommendations for EBM training, which can be used in the future to develop more structured teaching of the key EBM principles.¹⁴

Clinical Practice Guidelines

Busy clinicians may find it challenging to appraise literature for every clinical decision. Tools such as review articles and CPGs can be helpful to clinicians who wish to implement evidence-based practice.

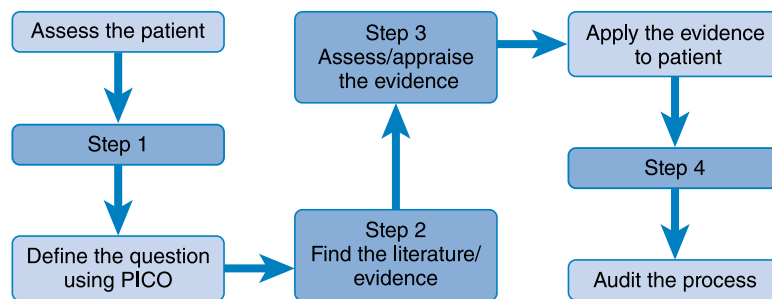
CPGs are systematically developed written statements made after a systematic evaluation of available evidence and intended to assist practitioners and patients in making appropriate decisions about health care in specific conditions or circumstances. Broadly speaking, CPGs aim to (1) describe most appropriate care based on the best available evidence, (2) reduce inappropriate variation in practice for a specified condition, (3) provide a more rational basis for interventions and referrals, (4) promote efficient use of health care resources, (5) act as a point of focus for quality control and audit, (6) highlight shortcomings of existing literature and suggest appropriate future research, and (7) provide an impetus for continuing education.

The schematic shown in Fig. 8.8 provides a model for how evidence-based guidelines are created. The IOM, in response to a request from the US Congress, formulated a report on the best methods to be used in developing CPGs that are rigorous and trustworthy. The IOM's eight standards for developing CPGs are considered the gold standard that every CPG aims to reach. CPGs critical to the field of PM&R that developed, advocated for, and were endorsed by the AAPM&R are available for review as well.⁶ Cochrane Rehabilitation is an online resource available for rehabilitation professionals, wherein Cochrane reviews and recommendations for clinical use on topics pertinent to rehabilitation are developed and available for use.¹⁹ Cochrane Rehabilitation proposes to help identify commonly used rehabilitation treatments that are well supported based on practice-based evidence; they do not require formal evidentiary support and offer a list of rehabilitation interventions that are evidence based despite lack of support from randomized controlled trials.⁴⁴

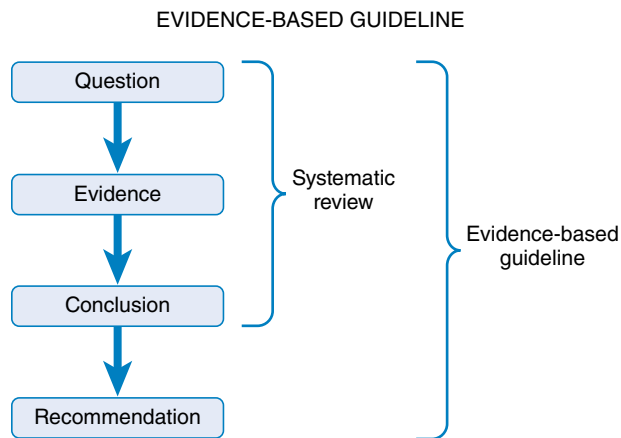
James M. Anderson Center for Health Systems Excellence has a division dedicated to support EBM decision making and provides topic-specific evidence-based care recommendations as CPGs relevant to pediatric care.³² CPGs relevant to management of patients with spinal cord injury, based on evidence-based research conducted by the North American Consortium for Spinal Cord Medicine, are published by the Paralyzed Veterans of America.⁴⁶

The Veterans Health Administration in collaboration with the Department of Defense has published guidelines that include management of upper and lower limb amputations and postacute mild traumatic brain injury. AAPM&R's Ethical Health Check Questionnaire committee reviews PM&R-relevant guidelines on a cyclic basis. At the time of this publication, the AAPM&R-endorsed guidelines include management of anterior cruciate ligament injuries, hip fractures, glenohumeral joint arthritis, among others.⁶

AAPM&R has called for a national plan to address the postacute sequelae of COVID-19 and form a multidisciplinary



• **Fig. 8.7** Incorporating evidence-based medicine into clinical practice. *PICO*, P: patient, problem, or population; I: intervention; C: comparison, control, or comparator; O: outcomes.



• **Fig. 8.8** Model for the creation of an evidence-based guideline.

collaborative of experts to provide clinical guidance to improve quality of care and provide resources and education for clinicians. The group has published a number of guidance statements on a rolling basis available on the AAPM&R website.⁵

Continuing Board Certification and Quality Improvement

The ABPMR is one of the 24-member boards of the American Board of Medical Specialties (ABMS). Since 1993, all certificates issued by the ABPMR were to be recertified every 10 years; in 2022, the process changed to a 5-year continuing certification cycle. The recertification process had been elevated to a more involved process of maintenance of certification before 2020 and now continuing certification. The change has taken place in an effort to assure the public that board-certified specialists have remained current with evolving knowledge, that their practices meet acceptable quality and safety standards, and that they are recognized and respected as specialists by their patients and peers. This ongoing process is designed to verify a diplomate's credentials, licensure, professional standing, and practice performance. Maintenance of and continuing certifications are intended to be quality drivers, but their relevance and value to society remain under scrutiny and are being challenged¹⁶ even today.

The ABPMR continuing certification program has one primary certification and six specialty certifications (directly offers brain injury medicine, pediatric rehabilitation medicine, and spinal cord injury medicine; cosponsors neuromuscular medicine, pain medicine, and sports medicine). For each of the three

cosponsored subspecialty certifications, another ABMS member board is responsible for examination development, administration, scoring, and analysis. The continuing certification program provides a mechanism for measuring, on a continual basis, the diplomate's competencies in the PM&R specialty. To maintain time-limited certification, diplomates are required to complete four components that measure the six competencies (Box 8.4).

The ABPMR states that the standards of certification are distinct from those of licensure. Possession of a board certificate does not indicate total qualification for practice privileges, nor does it imply exclusion of other physicians who are not so certified.

Diplomates' continuing certification participation status, either "meeting requirements" or "not meeting requirements" will be available to the public when using the searchable "Verify a Physician's Certification" tool on the ABPMR website.

Even though the maintenance of and continuing certification programs sound rational and generally seem well designed to advance medical knowledge and promote quality improvement among health care providers, the effectiveness, meaningfulness, and cost effectiveness of the recertification programs have not been proven scientifically.⁶² The eventual goal is that with greater transparency and accountability, a board-certified diplomate of any specialty will be held to equal standards. Through these programs, board-certified physicians are expected to advance the quality and standard of health care nationwide. The ABMS believes higher standards mean better care and considers the certification programs an essential element of this process. The ABPMR

• BOX 8.4 American Board of Physical Medicine and Rehabilitation Continuing Certification: Components and Competencies

Four Components

- Professionalism (professionalism and professional standing)
- Lifelong learning (category 1 continuing medical education credits and self-assessment)
- Assessment for learning (longitudinal assessment for physical medicine and rehabilitation)
- Quality improvement (quality improvement project)

Six Competencies

- Medical knowledge
- Patient care and procedural skills
- Interpersonal and communication skills
- Professionalism
- Practice-based learning and improvement
- Systems-based practice

supports its diplomates' dedication to lifelong learning and assures the public the quality of its diplomates.

Moving From the Maintenance of Certification to the Continuous Certification Program

According to the ABMS, diplomates have raised major concerns about maintenance of certification programs. Recognizing that a comprehensive revision to this certification process was required, the ABMS has reviewed both the framework and purpose of continuing certification of physicians and launched a process, entitled *Continuing Board Certification: Vision for the Future (The Vision Initiative)*.⁸ After extensive research done by *The Vision Initiative* Commission of ABMS, the ABMS has recommended continuance of the certification program with a robust, evidence-based, efficient, and fair approach. The name maintenance of certification has been replaced by the newer term, continuing certification program.

To address similar concerns raised by its diplomates, the ABPMR implemented in January 2020 the Longitudinal Assessment for PM&R (LA-PM&R), representing a novel approach to ABPMR certification that is more focused on learning and continuous improvement. According to the ABPMR, the LA-PM&R will:

1. Replace the secure maintenance of certification part III examination
2. Count for self-assessment continuing medical education (SA-CME) requirements (earn up to 12 SA-CME credits per year of LA-PM&R participation)
3. Be part of your 5-year continuing certification program requirements, which also include professionalism, lifelong learning (CME), and quality improvement
4. Ask 15 to 20 questions per quarter
5. Provide customized content based on five domains
6. Provide immediate feedback and learning resources

Participants can customize the LA-PM&R content by adjusting the relative percentages of the following categories:

- Central nervous system rehabilitation: brain injury, spinal cord injury, and stroke

- Musculoskeletal and pain medicine: arthritis, acute/chronic trauma, fractures, spinal disorders, and sports medicine
- Neuromuscular medicine/electrodiagnosis: electromyography, neuropathies, nerve conduction studies, neuromuscular transmission
- Medical rehabilitation: amputation, cancer, cardiopulmonary, infectious disease, genitourinary/gastrointestinal, and transplant
- Pediatric rehabilitation medicine: related topics

LA-PM&R is open to all diplomates as part of the continuing certification program in PM&R as of January 2020.^{11,12}

Quality Improvement Projects and Project Options

As one of six competencies of ABPMR, practice-based learning and improvement is addressed by the continuing certification program. Its diplomates must demonstrate the ability to investigate and evaluate the care of patients, appraise and assimilate scientific evidence, and continuously improve patient care based on constant self-evaluation and lifelong learning. Diplomates with time-limited certificates issued between 2012 and 2019 must complete two ABPMR-approved practice performance projects (one project in years 1–5 and one project in years 6–10) during the 10-year maintenance of certification cycle. Diplomates with certificates issued in 2020 and beyond must complete one ABPMR-approved quality improvement project during each 5-year continuing certification cycle (Table 8.2).¹²

The ABPMR suggested three common options before year 2020 for fulfilling the practice improvement project requirement, and since then three more options have been added.¹⁰ The term *practice improvement project* has been replaced with *quality improvement project*. There are several quality improvement project options available to PM&R physicians through various professional organizations (see later).

ABPMR Self-Directed Quality Improvement Project

The ABPMR has developed a quality improvement project wizard that guides diplomates to pertinent quality improvement project options based on their self-identified clinical areas of practice.⁹

TABLE 8.2 Activities Required to Recertify

Certificate Issue Date	Activities Required to Meet Requirements for Continuing Certification (CC)
2012–2019 (10-year MOC cycle)	• Licensure • Complete and report 150 Category 1 CME credits in years 1–5 and 6–10 of the MOC cycle (for a total of 300 Category 1 CME credits) • Complete an average of 8 CME credits per year (for a total of 40 CME credits in years 1–5 and 40 CME credits in years 6–10) involving ABPMR-approved self-assessments • Examination: if certification expires in 2020 or later, will enter into a continuing certification cycle, including LA-PM&R assessment • Complete two ABPMR-approved PIPs (one in years 1–5 and one in years 6–10)
2020 and beyond (5-year CC cycle)	• Licensure • Complete and report 150 Category 1 CME credits in 5 years (including 50 SA-CME), which is on average 30 CME credits annually • Complete 50 ABPMR-approved self-assessment CME credits as previously mentioned • LA-PM&R assessment: once a quarter test through an online platform from years 1–4, and a summative assessment in year 5. The lowest four quarters of the 16 quarters are dropped from scoring (from year 1–4). Earns 3 SA-CME per quarter • Complete 1 ABPMR-approved QI project

ABPMR, American Board of Physical Medicine and Rehabilitation; CC, continuing certification; CME, continuing medical education; LA, longitudinal assessment; MOC, maintenance of certification; PM&R, physical medicine and rehabilitation; PIP, practice improvement project; QI, quality improvement; SA, self-assessment.

The wizard offers more than 20 projects for users to choose that encompass quality improvements not only in clinical areas but also in such things as administration, education, and research. The self-directed quality improvement project allows individual diplomates the opportunity to demonstrate improvement within their own practice using a continuous quality improvement process and PDSA methodology. This option is free of charge.

American Academy of Physical Medicine and Rehabilitation Preformatted Quality Improvement Project

AAPM&R offers four practice improvement projects on deep vein thrombosis, stroke rehabilitation, osteoporosis, and low back pain. After deciding on a project, the diplomate will receive a toolkit to guide the process. The diplomate starts the process by answering a series of set questions: self-evaluation of an issue, practice-based assessment of the issue, and self-selection of three areas for improvement from a predetermined list based on answers to the self-evaluation and practice-based assessment questions. All diplomates create an improvement program specific to their practice with a variety of tools and resources available.

The non-AAPM&R member diplomate must pay a fee for this service and earn 20 CME credit hours. Minimum 30-day period of data collection and analysis is recommended.

American Association of Neuromuscular and Electrodiagnostic Medicine Performance in Practice

The American Association of Neuromuscular and Electrodiagnostic Medicine (AANEM) is an organization designed to enhance quality patient care by focusing on providing quality electrodiagnostic (EDX) services. It has three performance-in-practice projects available: ulnar neuropathy, carpal tunnel syndrome, and EDX report writing. Each project was formed from an expert consensus document created by AANEM members. According to AANEM, the purpose of the quality improvement projects is to enable a physician to apply best practices in the electrodiagnosis of ulnar neuropathy, carpal tunnel syndrome, or EDX report writing. Each module will help physicians evaluate the quality of their EDX practice on the specific topic and identify areas of potential improvement and implement the action for changes.

A non-AANEM diplomate pays a fee for this service and earns 20 CME credit hours. A minimal 30-day period of data collection and analysis is recommended.

American Board of Medical Specialties Portfolio Program

The ABMS Portfolio Program works with many types of health care organizations to recognize the work physicians are already doing to improve their practices and the care of their patients. The sponsors (health care organizations) support physician involvement in local improvement initiatives by awarding credit for continuing certification from a participating ABMS member board, including ABPMR. Only a diplomate whose affiliated organization is on the ABMS sponsor list can use this program for the quality improvement credit.

National Committee on Quality Assurance Patient-Centered Specialty Practice Recognition

The NCQA Patient-Centered Specialty Practice Recognition program builds on the success of the PCMH recognition program by recognizing specialty practices that excel in delivering high-quality, patient-centered care. It focuses on proactive coordination and sharing of information. Everyone in the practice works as a team to coordinate care with primary care, other referring

clinicians, community resources, and secondary services. This project is done at an organizational level with purchased recognition service from NCQA.

The American Institute of Ultrasound in Medicine Practice Accreditation in Musculoskeletal Ultrasound-Guided Interventional Procedures

The American Institute of Ultrasound in Medicine (AIUM) ultrasound practice accreditation allows practices to demonstrate that they meet or exceed nationally recognized standards in the performance and interpretation of diagnostic ultrasound examinations. A diplomate needs to submit both an application and case study documents to be evaluated for accreditation. The accreditation process is a paid service. ABPMR accepts the AIUM practice accreditation in musculoskeletal ultrasound-guided interventional procedure as a quality improvement project credit.

Summary

With a team-based, patient-centric, health care quality mission of safe, efficient, cost-effective, and equitable health care for all, rehabilitation professionals, including physiatrists, must work together to achieve best care quality and outcomes for patients. Practicing safe EBM that is focused on value and outcomes can help us get there. Fair, transparent, valid, and objective measurements of quality and safety should be embraced. It is together that we need to continue our journey to an accessible, affordable, high-quality health care system that is valuable and sustainable over the long term for all.

Key References

- American Academy of Physical Medicine and Rehabilitation. *Merit-Based Incentive Payment System (MIPS)*. <https://www.aapmr.org/quality-practice/quality-reporting/quality-payment-program/merit-incentive-payment-system>
- American Academy of Physical Medicine and Rehabilitation. *PASC Consensus Guidance*. <https://www.aapmr.org/advocacy/current-priorities/long-covid-pasc/pasc-guidance>
- American Board of Medical Specialties. *Continuing Board Certification: Vision for the Future Commission Final Report*, February 12, 2019. ABMS; 2019.
- American Board of Physical Medicine and Rehabilitation. *8 Steps to Complete a QI Project*. Accessed July 10, 2024. <https://www.abpmr.org/CC/QualityImprovement/SelfDirected>
- American Board of Physical Medicine and Rehabilitation. *Improving Your Medical Practice: Demonstrating Quality*. Accessed July 10, 2024. <https://www.abpmr.org/CC/QualityImprovement>
- American Board of Physical Medicine and Rehabilitation. *Looking Forward to 2020: A New Era in Certification*. <https://www.abpmr.org/NewsCenter/Detail/2020-new-era-certification>
- American Board of Physical Medicine and Rehabilitation. *WHY [Continuing Certification] CC? Learning Over a Lifetime*. Accessed July 10, 2024. <https://www.abpmr.org/CC/About/Overview>
- Annaswamy T, Kasinton D, Royston A. Value-based care and musculoskeletal rehabilitation. *Curr Phys Med Rehabil Rep*. 2018;6(1):49-54.
- Annaswamy TM, Rizzo JR, Schnappinger A, et al. Evidence-based medicine training in United States-based physiatry residency programs. *Am J Phys Med Rehabil*. 2022;101(7/1):S40-S44.

15. CARF International. *ASPIRE to Excellence Quality Framework*. 2024. <https://carf.org/accreditation/our-standards/>
16. Change Board Recertification. *Anti-MOC*. 2024. <http://www.changeboardrecert.com/anti-moc.html>.
17. Chassin MR. Is health care ready for Six Sigma quality? *Milbank Q*. 1998;76(4):565-591.
18. Chassin MR, Loeb JM, Schmaltz SP, et al. Accountability measures—using measurement to promote quality improvement. *N Engl J Med*. 2010;363(7):683-688.
20. Co JPT, Weiss KB, CLER Evaluation Committee. CLER pathways to excellence, version 2.0: executive summary. *J Grad Med Educ*. 2019;11(6):739-741.
21. Connor L, Dean J, McNett M, et al. Evidence-based practice improves patient outcomes and healthcare system return on investment: findings from a scoping review. *Worldviews Evid Based Nurs*. 2023;20(1):6-15.
22. Donabedian A. The quality of care. How can it be assessed? *JAMA*. 1988;260(12):1743-1748.
23. Evidence-Based Medicine Working Group. Evidence-based medicine. A new approach to teaching the practice of medicine. *JAMA*. 1992;268(17):2420-2425.
24. Frye AW, Hemmer PA. Program evaluation models and related theories: AMEE guide no. 67. *Med Teach*. 2012;34(5):e288-e299.
25. Glasziou P, Guyatt GH, Dans AL, et al. Applying the results of trials and systematic reviews to individual patients. *ACP J Club*. 1998;129(3):A15-A16.
26. Grover P, Volshteyn O, Carr DB. Physical medicine and rehabilitation residency quality improvement and research curriculum: design and implementation. *Am J Phys Med Rehabil*. 2021;100(2/1):S23-S29.
27. Hagland M. *Transformative Quality: The Emerging Revolution in Health Care Performance*. CRC Press; 2008.
28. HealthMeasures. *PROMIS [Patient-Reported Outcomes Measurement Information System]: Dynamic Tools to Measure Health Outcomes from the Patient Perspective*. National Institutes of Health. Accessed June 25, 2014. <http://www.nihpromis.org>
30. Institute of Medicine. *Crossing the Quality Chasm: A New Health System for the 21st Century*. National Academies Press; 2001.
31. Institute of Medicine (US) Committee on Data Standards for Patient Safety, Aspden P, Corrigan JM, et al., eds. *Patient Safety: Achieving a New Standard for Care*. National Academies Press; 2004.
33. Jette AM. Toward a common language for function, disability, and health. *Phys Ther*. 2006;86(5):726-734.
34. Jette AM, Haley SM. Contemporary measurement techniques for rehabilitation outcomes assessment. *J Rehabil Med*. 2005;37(6):339-345.
35. Jette AM, Tao W, Haley SM. Blending activity and participation sub-domains of the ICF. *Disabil Rehabil*. 2007;29(22):1742-1750.
36. Kind AJ, Smith MA, Liou JI, et al. Discharge destination's effect on bounce-back risk in Black, White, and Hispanic acute ischemic stroke patients. *Arch Phys Med Rehabil*. 2010;91(2):189-195.
38. Mainz J. Defining and classifying clinical indicators for quality improvement. *Int J Qual Health Care*. 2003;15(6):523-530.
39. Martinkewiz P. Commission on the Accreditation of Rehabilitation Facilities: steps for success. *Rehabil Nurs*. 2023;48(4):133-136.
40. Medicare.GOV. *Hospital Compare*. Accessed July 13, 2024. <http://www.medicare.gov/hospitalcompare/search.html?AspxAutoDetectCookieSupport=1>
41. Middleton A, Graham JE, Krishnan S, et al. Program interruptions and short-stay transfers represent potential targets for inpatient rehabilitation care-improvement efforts. *Am J Phys Med Rehabil*. 2016;95(11):850-861.
42. National Committee for Quality Assurance. *About*. Accessed February 26, 2019. <https://www.ncqa.org/about-ncqa/>; https://www.qualityforum.org/Infographics/Lifecycle_of_Performance_Measures.aspx
44. Negrini S. Evidence in rehabilitation medicine: between facts and prejudices. *Am J Phys Med Rehabil*. 2019;98(2):88-96.
48. Porter ME. What is value in health care? *N Engl J Med*. 2010;363(26):2477-2481.
49. Raphael J, Sadof M, Stille C, et al. *Not Just Little Adults: Considerations for Quality Measures of Child Health Care*. National Quality Measures Clearinghouse Expert Commentaries; 2014.
50. Rivara FP, Ennis SK, Mangione-Smith R, et al. Quality of care indicators for the rehabilitation of children with traumatic brain injury. *Arch Phys Med Rehabil*. 2012;93(3):381-385.e9.
51. Sackett DL, Straus S, Richardson W, et al. *Evidence-Based Medicine: How to Practice and Teach EBM*. 2nd ed. Churchill Livingstone; 2000.
52. Sackett DL, Rosenberg WM, Gray JA, et al. Evidence based medicine: what it is and what it isn't. *BMJ*. 1996;312(7023):71-72.
53. Shaughnessy AF, Slawson DC. What happened to the valid POEMs? A survey of review articles on the treatment of type 2 diabetes. *BMJ*. 2003;327(7409):266.
55. Stratford PW, Binkley JM. A comparison study of the back pain functional scale and Roland Morris Questionnaire. North American Orthopaedic Rehabilitation Research Network. *J Rheumatol*. 2000;27(8):1928-1936.
56. Stratford PW, Binkley JM, Riddle DL. Development and initial validation of the back pain functional scale. *Spine*. 2000;25(16):2095-2102.
59. US Centers for Medicare & Medicaid Services. *Functional Measures*. 2023. <https://www.cms.gov/medicare/quality/initiatives/pac-quality-initiatives/functional-measures>
60. US Centers for Medicare & Medicaid Services. *Publication of Potentially Preventable Readmission Measures for the SNF, IRF, LTCH, and HH Quality Reporting Programs (QRPs)—Questions and Answers*. May 2019. <https://www.hhs.gov/guidance/document/publication-potentially-preventable-readmission-measures-snf-irf-ltch-and-hh-qrrps-q-may>
61. US Centers for Medicare & Medicaid Services. *Quality Payment Program*. 2024. <https://qpp.cms.gov/>
62. Van Harrison R, Olson CA. Evolving health care systems and approaches to maintenance of certification. *J Contin Educ Health Prof*. 2013;33(1):S1-S4.
63. Welch HG, Lurie JD. Teaching evidence-based medicine: caveats and challenges. *Acad Med*. 2000;75(3):235-240.
64. Wright JG, Swiontkowski MF, Heckman JD. Introducing levels of evidence to the journal. *J Bone Joint Surg Am*. 2003;85-a(1):1-3.
65. Zakowski L, Seibert C, VanEyck S. Evidence-based medicine: answering questions of diagnosis. *Clin Med Res*. 2004;2(1):63-69.

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References

- Agency for Healthcare Research and Quality. *Guidelines and Measures*, 2014. Accessed April 9, 2025. <https://www.ahrq.gov/gam/index.html>
- Agency for Healthcare Research and Quality. *Pediatric Quality Indicator Measures*, 2024. https://qualityindicators.ahrq.gov/measures/pdi_resources
- American Academy of Physical Medicine and Rehabilitation. *2019 Measures*. Accessed February 27, 2019. <https://www.aapmr.org/quality-practice/registry/measures/2019-measures>
- American Academy of Physical Medicine and Rehabilitation. *Merit-Based Incentive Payment System (MIPS)*. <https://www.aapmr.org/quality-practice/quality-reporting/quality-payment-program/merit-incentive-payment-system>
- American Academy of Physical Medicine and Rehabilitation. *PASC Consensus Guidance*. <https://www.aapmr.org/advocacy/current-priorities/long-covid-pasc/pasc-guidance>
- American Academy of Physical Medicine and Rehabilitation. *PM&R Applicable Guidelines*. Accessed February 26, 2019. <https://www.aapmr.org/quality-practice/clinical-practice-guidelines/aapm-r-endorsed-guidelines>
- American Academy of Physical Medicine and Rehabilitation. *Registry Measures*. Accessed February 27, 2019. <https://www.aapmr.org/quality-practice/registry/measures>
- American Board of Medical Specialties. *Continuing Board Certification: Vision for the Future Commission Final Report, February 12, 2019*. ABMS; 2019.
- American Board of Physical Medicine and Rehabilitation. *8 Steps to Complete a QI Project*. Accessed July 10, 2024. <https://www.abpmr.org/CC/QualityImprovement/SelfDirected>
- American Board of Physical Medicine and Rehabilitation. *Improving Your Medical Practice: Demonstrating Quality*. Accessed July 10, 2024. <https://www.abpmr.org/CC/QualityImprovement>
- American Board of Physical Medicine and Rehabilitation. *Looking Forward to 2020: A New Era in Certification*. <https://www.abpmr.org/NewsCenter/Detail/2020-new-era-certification>
- American Board of Physical Medicine and Rehabilitation. *WHY [Continuing Certification] CC? Learning Over a Lifetime*. Accessed July 10, 2024. <https://www.abpmr.org/CC/About/Overview>
- Annaswamy T, Kasitinon D, Royston A. Value-based care and musculoskeletal rehabilitation. *Curr Phys Med Rehabil Rep*. 2018;6(1):49-54.
- Annaswamy TM, Rizzo JR, Schnappinger A, et al. Evidence-based medicine training in United States-based psychiatry residency programs. *Am J Phys Med Rehabil*. 2022;101(7/1):S40-S44.
- CARF International. *ASPIRE to Excellence Quality Framework*. 2024. <https://carf.org/accreditation/our-standards/>
- Change Board Recertification. *Anti-MOC*. 2024. <http://www.changeboardrecert.com/anti-moc.html>
- Chassin MR. Is health care ready for Six Sigma quality? *Milbank Q*. 1998;76(4):565-591.
- Chassin MR, Loeb JM, Schmaltz SP, et al. Accountability measures—using measurement to promote quality improvement. *N Engl J Med*. 2010;363(7):683-688.
- Chochrane Rehabilitation. *Mission and Goals*. Accessed February 27, 2019. <https://rehabilitation.cochrane.org/about-us/mission-and-goals>
- Co JPT, Weiss KB, CLER Evaluation Committee. CLER pathways to excellence, version 2.0: executive summary. *J Grad Med Educ*. 2019;11(6):739-741.
- Connor L, Dean J, McNett M, et al. Evidence-based practice improves patient outcomes and healthcare system return on investment: findings from a scoping review. *Worldviews Evid Based Nurs*. 2023;20(1):6-15.
- Donabedian A. The quality of care. How can it be assessed? *JAMA*. 1988;260(12):1743-1748.
- Evidence-Based Medicine Working Group. Evidence-based medicine. A new approach to teaching the practice of medicine. *JAMA*. 1992;268(17):2420-2425.
- Frye AW, Hemmer PA. Program evaluation models and related theories: AMEE guide no. 67. *Med Teach*. 2012;34(5):e288-e299.
- Glasziou P, Guyatt GH, Dans AL, et al. Applying the results of trials and systematic reviews to individual patients. *ACP J Club*. 1998;129(3):A15-A16.
- Grover P, Volshteyn O, Carr DB. Physical medicine and rehabilitation residency quality improvement and research curriculum: design and implementation. *Am J Phys Med Rehabil*. 2021;100(2/1):S23-S29.
- Hagland M. *Transformative Quality: The Emerging Revolution in Health Care Performance*. CRC Press; 2008.
- HealthMeasures. *PROMIS [Patient-Reported Outcomes Measurement Information System]: Dynamic Tools to Measure Health Outcomes From the Patient Perspective*. National Institutes of Health. Accessed June 25, 2014. <http://www.nihpromis.org>
- IBM. *What Are Healthcare Performance Measurements?* 2024. <https://www.ibm.com/topics/healthcare-performance-measurements>
- Institute of Medicine. *Crossing the Quality Chasm: A New Health System for the 21st Century*. National Academies Press; 2001.
- Institute of Medicine (US) Committee on Data Standards for Patient Safety, Aspden P, Corrigan JM, et al., eds. *Patient Safety: Achieving a New Standard for Care*. National Academies Press; 2004.
- James M. Anderson Center for Health Systems Excellence. *Evidence-Based Care Guidelines*. Cincinnati Children's; 2024. <https://www.cincinnatichildrens.org/research/divisions/j/anderson-center/evidence-based-care/recommendations>
- Jette AM. Toward a common language for function, disability, and health. *Phys Ther*. 2006;86(5):726-734.
- Jette AM, Haley SM. Contemporary measurement techniques for rehabilitation outcomes assessment. *J Rehabil Med*. 2005;37(6):339-345.
- Jette AM, Tao W, Haley SM. Blending activity and participation sub-domains of the ICF. *Disabil Rehabil*. 2007;29(22):1742-1750.
- Kind AJ, Smith MA, Liou JI, et al. Discharge destination's effect on bounce-back risk in Black, White, and Hispanic acute ischemic stroke patients. *Arch Phys Med Rehabil*. 2010;91(2):189-195.
- Leapfrog Hospital Safety Grade. *How Safe is Your Hospital?* The Leapfrog Group; 2024. <https://www.hospitalsafetygrade.org/>
- Mainz J. Defining and classifying clinical indicators for quality improvement. *Int J Qual Health Care*. 2003;15(6):523-530.
- Martinkewiz P. Commission on the Accreditation of Rehabilitation Facilities: steps for success. *Rehabil Nurs*. 2023;48(4):133-136.
- Medicare.GOV. *Hospital Compare*. Accessed July 13, 2024. <http://www.medicare.gov/hospitalcompare/search.html?AspxAutoDetectCookieSupport=1>
- Middleton A, Graham JE, Krishnan S, et al. Program interruptions and short-stay transfers represent potential targets for inpatient rehabilitation care-improvement efforts. *Am J Phys Med Rehabil*. 2016;95(11):850-861.
- National Committee for Quality Assurance. *About*. Accessed February 26, 2019. <https://www.ncqa.org/about-ncqa/>; https://www.qualityforum.org/Infographics/Lifecycle_of_Performance_Measures.aspx
- National Quality Forum. *An Illustrative Example. Lifecycle of a Performance Measure: Depression Remission at 6 months*. October, 2012. Accessed February 26, 2019. https://www.qualityforum.org/Infographics/Lifecycle_of_Performance_Measures.aspx
- Negrini S. Evidence in rehabilitation medicine: between facts and prejudices. *Am J Phys Med Rehabil*. 2019;98(2):88-96.
- Oxford Centre for Evidence-Based Medicine. *The Oxford 2011 Levels of Evidence*. <http://www.cebm.net/index.aspx?o=5653>
- Paralyzed Veterans of America. *Clinical Practice Guidelines: Spinal Cord Medicine*. Paralyzed Veterans of America; 2024. <https://pva.org/research-resources/publications/clinical-practice-guidelines/>
- PCPI Foundation. *Measure Science Practice Areas*. Accessed February 26, 2019. <https://www.thepcpi.org/page/MeasureAreas?&hsearchterms=%22measure+and+testing+and+protocol%22>
- Porter ME. What is value in health care? *N Engl J Med*. 2010;363(26):2477-2481.

49. Raphael J, Sadof M, Stille C, et al. *Not Just Little Adults: Considerations for Quality Measures of Child Health Care*. National Quality Measures Clearinghouse Expert Commentaries; 2014.
50. Rivara FP, Ennis SK, Mangione-Smith R, et al. Quality of care indicators for the rehabilitation of children with traumatic brain injury. *Arch Phys Med Rehabil*. 2012;93(3):381-385.e9.
51. Sackett DL, Straus S, Richardson W, et al. *Evidence-Based Medicine: How to Practice and Teach EBM*. 2nd ed. Churchill Livingstone; 2000.
52. Sackett DL, Rosenberg WM, Gray JA, et al. Evidence based medicine: what it is and what it isn't. *BMJ*. 1996;312(7023):71-72.
53. Shaughnessy AF, Slawson DC. What happened to the valid POEMs? A survey of review articles on the treatment of type 2 diabetes. *BMJ*. 2003;327(7409):266.
54. Shirley Ryan AbilityLab. *Rehabilitation Measures Database*. Accessed February 26, 2019. <https://www.sralab.org/rehabilitation-measures>
55. Stratford PW, Binkley JM. A comparison study of the back pain functional scale and Roland Morris Questionnaire. North American Orthopaedic Rehabilitation Research Network. *J Rheumatol*. 2000; 27(8):1928-1936.
56. Stratford PW, Binkley JM, Riddle DL. Development and initial validation of the back pain functional scale. *Spine*. 2000;25(16): 2095-2102.
57. US Centers for Medicare & Medicaid Services. *The Continuity Assessment Record and Evaluation (CARE) Item Set and B-CARE*. 2023. Accessed April 9, 2025. <https://www.cms.gov/medicare/quality/initiatives/pac-quality-initiatives/care-item-set-and-b-care>
58. US Centers for Medicare & Medicaid Services. *The Development and Testing of the Continuity Assessment Record and Evaluation (CARE) Item Set: Final Report on CARE Item Set and Current Assessment Comparisons*. vol 3. RTI International; 2012.
59. US Centers for Medicare & Medicaid Services. *Functional Measures*. 2023. <https://www.cms.gov/medicare/quality/initiatives/pac-quality-initiatives/functional-measures>
60. US Centers for Medicare & Medicaid Services. *Publication of Potentially Preventable Readmission Measures for the SNF, IRE, LTCH, and HH Quality Reporting Programs (QRPs)—Questions and Answers*. May 2019. <https://www.hhs.gov/guidance/document/publication-potentially-preventable-readmission-measures-snf-irf-ltch-and-hh-qtps-q-may>
61. US Centers for Medicare & Medicaid Services. *Quality Payment Program*. 2024. <https://qpp.cms.gov/>
62. Van Harrison R, Olson CA. Evolving health care systems and approaches to maintenance of certification. *J Contin Educ Health Prof*. 2013;33(1):S1-S4.
63. Welch HG, Lurie JD. Teaching evidence-based medicine: caveats and challenges. *Acad Med*. 2000;75(3):235-240.
64. Wright JG, Swiontkowski MF, Heckman JD. Introducing levels of evidence to the journal. *J Bone Joint Surg Am*. 2003;85-a(1):1-3.
65. Zakowski L, Seibert C, VanEyck S. Evidence-based medicine: answering questions of diagnosis. *Clin Med Res*. 2004;2(1):63-69.